

Logical Address and Physical address

Example:

Suppose a logical address space of 32 pages of 2048 words. There are 64 frames in the main memory. According to given information calculate the various parameters related to paging.

Calculate the size of Logical Address? Calculate the size of Physical Address?

logical address contains: page number and page offset (page size)

physical address contains : number of frames and frame size

Logical address space contains 32 pages and to represent 32 pages we need 5 bits

$$2^5 = 32$$

Each page contains 2048 words therefore the offset or "d" parameter can be represented by 11 bits $2^{11} = 2048$.

So total number of bits to represent logical address : $5 + 11 = 16$ bits.

Similarly 64 frames can be represented by 6 bits $2^6 = 64$ and offset will be same (because page size is equal to frame size) i.e. 11 bits. The total number of bits to represent physical address: $6 + 11 = 17$ bits.